

The Szeged–Wiener difference of a 2-connected graph is at least $\min\{2n, 3n - 10\}$

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Abstract

For every 2-connected graph G of order $n \geq 4$ other than K_n, K_n^2, K_n^{n-2} , we prove

$$\text{Sz}(G) - W(G) \geq \min\{2n, 3n - 10\}.$$

This proves Conjecture 5 of Bonamy, Knor, Lužar, Pinlou and Škrekovski for $n \geq 10$, and both branches of the stronger bound are sharp. The proof uses an exact deletion identity for dominated vertices, a structural description of vertices with small good-edge contribution, and induction.

This note was produced by AI within the [Graph Theory AI](#) project: the argument was found by a language model and audited by an adversarial LLM referee, and it has not been checked by a human mathematician. [Section 5](#) records the full provenance.

1 Introduction

For a connected graph G , let

$$W(G) = \sum_{\{a,b\} \subseteq V(G)} d(a,b), \quad \text{Sz}(G) = \sum_{uv \in E(G)} n_{uv}(u)n_{uv}(v),$$

where $n_{uv}(u)$ counts the vertices closer to u than to v , and put $\eta(G) = \text{Sz}(G) - W(G)$. Following [1], K_n^t is obtained from K_{n-1} by adding a vertex adjacent to exactly t old vertices. The source paper proves $\eta(G) \geq 2n - 6$ for every 2-connected noncomplete G , with equality exactly for K_n^2, K_n^{n-2} , and asks:

Conjecture 1.1 ([1, Conjecture 5]). Let G be a 2-connected graph of order $n \geq 10$ not isomorphic to K_n, K_n^2 , or K_n^{n-2} . Then $\eta(G) \geq 2n$.

An edge uv is *good* for $\{a, b\}$ if, after possibly exchanging u, v , $d(a, u) < d(a, v)$ and $d(b, v) < d(b, u)$. This terminology is due to Simić, Gutman and Baltić [2]. Write $g_G(a, b)$ for the number of good edges and

$$D_G(a, b) = g_G(a, b) - d_G(a, b), \quad c_G(a) = \sum_{b \neq a} D_G(a, b).$$

Every shortest-path edge is good, so $D_G(a, b) \geq 0$. Proposition 6 of [1] gives

$$\eta(G) = \sum_{\{a,b\}} D_G(a, b) = \frac{1}{2} \sum_a c_G(a). \tag{1}$$

Our exact dominated-vertex formula strengthens [1, Lemma 7].

Theorem 1.2 (Main theorem). *Let G be a 2-connected graph of order $n \geq 4$, with $G \not\cong K_n, K_n^2, K_n^{n-2}$. Then*

$$\eta(G) \geq \min\{2n, 3n - 10\}.$$

Both branches are sharp for every applicable $n \geq 5$.

Idea of proof. If $\eta(G) < 2n$, (1) gives a vertex of contribution at most three. A breadth-first-layer analysis produces, apart from C_5 , a dominated vertex u whose deletion remains 2-connected. An exact deletion identity shows that removing u loses at least four units of η . Induction applies unless $G - u$ is excluded; those predecessors are handled directly. Graphs with a universal vertex satisfy $3n - 10$ by counting induced P_3, P_4, C_4 . The extremal families appear in Section 4. [\[full proof\]](#)

Corollary 1.3. *Conjecture 1.1 holds.*

Proof. For $n \geq 10$, $3n - 10 \geq 2n$, so Theorem 1.2 gives $\eta(G) \geq 2n$. □

2 The deletion mechanism

Lemma 2.1 (Dominated-vertex deletion). *Suppose G is 2-connected, $|V(G)| \geq 4$, and $N_G[u] \subseteq N_G[v]$ for distinct u, v . Put $H = G - u$. Then:*

- (i) H is isometric in G ;
- (ii) the only possible cut vertex of H is v ;
- (iii) if $q_G(u)$ counts incidences $(\{a, b\}, e)$ in which $a, b \in V(H)$ and e is incident with u and good for $\{a, b\}$, then $\eta(G) - \eta(H) = c_G(u) + q_G(u)$;
- (iv) $q_G(u) = 0$ when u is simplicial, while $q_G(u) \geq 2$ otherwise.

Idea of proof. Any detour xuy can be replaced by a walk through v , so old distances do not change. The difference in η splits into pairs containing u and old pairs whose new good edge meets u . A non-simplicial u has two nonadjacent neighbours, and both incident edges are good for their pair. [\[full proof\]](#)

Lemma 2.2 (Low-contribution structure). *Let G be 2-connected and let a be non-universal with $c_G(a) \leq 3$. Every vertex lies at distance at most two from a , and one of the following holds, where $A = N(a)$.*

- I. *There are adjacent $p, q \in A$, a vertex $z \notin N[a]$ with $N(z) \cap A = \{p, q\}$, and a set X such that $V(G) \setminus N[a] = \{z\} \dot{\cup} X$. Either X is empty, or $G[X]$ is connected, every $x \in X$ has $N(x) \cap A = \{p\}$, and exactly one edge joins z to X .*
- II. *There are distinct $p, q \in A$ and nonempty X, Y partitioning $V(G) \setminus N[a]$ such that $G[X], G[Y]$ are connected, $N(x) \cap A = \{p\}$ for $x \in X$, $N(y) \cap A = \{q\}$ for $y \in Y$, and exactly one edge xy joins X to Y .*

Moreover every non-universal vertex of a 2-connected graph satisfies $c_G(a) \geq 2$.

Idea of proof. Use distance layers from a . A vertex with two shortest paths contributes at least two, and two such vertices exceed the budget three. If shortest paths are unique, a same-layer edge joining different predecessor classes forces two contributions, leaving exactly one and hence Type II. If one vertex has two shortest paths, it has exactly two adjacent predecessors and at most one same-layer neighbour, giving Type I. Deeper layers or extra components create a cut vertex. [\[full proof\]](#)

Proposition 2.3 (Four-unit reduction). *Let G be 2-connected, have no universal vertex, and satisfy $\eta(G) < 2|V(G)|$. Then either $G \cong C_5$, or there is a vertex u dominated by another vertex such that $G - u$ is 2-connected and $\eta(G) - \eta(G - u) \geq 4$.*

Idea of proof. Choose a with $c_G(a) \leq 3$, apply [Lemma 2.2](#), and delete a suitable non-cut vertex in a connected Type I/II piece. For non-simplicial u , [Lemma 2.1](#) gives $c_G(u) + q_G(u) \geq 2 + 2$. If u is simplicial and $c_G(u) \leq 3$, Type II contains an isometric C_4 and every remaining vertex contributes two more units, forcing $\eta(G) \geq 2|V(G)|$, a contradiction. [\[full proof\]](#)

3 Universal vertices

Let $p_3(Q), p_4(Q), c_4(Q)$ count induced P_3, P_4, C_4 .

Proposition 3.1 (Universal-vertex estimate). *If G is 2-connected of order n , has a universal vertex, and is not K_n, K_n^2, K_n^{n-2} , then $\eta(G) \geq 3n - 10$.*

Idea of proof. For every diameter-two graph,

$$\eta(G) = 2(p_3(G) - \overline{m}(G)) + p_4(G) + 4c_4(G).$$

For the cone $G = K_1 \vee Q$, this becomes $\eta(G) = 2p_3(Q) + p_4(Q) + 4c_4(Q)$. Layering connected Q from a maximum clique gives $3|Q| - 7$, except for $K_{|Q|} - e$ and a clique with a pendant vertex, which give the two excluded noncomplete cones. [\[full proof\]](#)

4 Sharpness, checks, and status

Remark 4.1 (Sharpness). For $G = K_1 \vee P_{n-1}$, the cone identity gives $\eta(G) = 2(n - 3) + (n - 4) = 3n - 10$. For the second family, take a clique of order $n - 2$, an additional edge xy , and two cross edges forming a matching. Beginning with its induced C_4 , every added clique vertex is simplicial and contributes two, so $\eta(G) = 8 + 2(n - 4) = 2n$. We do not claim these are the only equality graphs.

Remark 4.2 (Referee computations). The referee exhaustively checked every 2-connected graph through order 11. For $n = 5, \dots, 11$, the admissible minima were 5, 8, 11, 14, 17, 20, 22, exactly the theorem's bound; the $n = 11$ run comprised 900,969,091 graphs. Every lemma passed on all 194,066 graphs of order 9 and a sample of 820,139 graphs of order 10. Further searches used annealing through order 24, exhaustive large-clique families through 14, structured families through 30, and all one-, two-, and three-edge perturbations of the extremal examples at orders 12, 13; none violated the claim. These computations support but are not used by the proof.

Remark 4.3 (Priority risk). Zhijie He's manuscript *The Strengthened Szeged–Wiener Conjecture* was posted on 8 September 2026 [\[3\]](#), nine days before this argument was generated. It claims the same theorem by a related dominated-vertex strategy, but has no arXiv identifier, DOI, or venue, and the associated MathDB problem labels it incomplete and unverified. A human must compare the arguments before making any novelty or priority claim. The referee found no published or arXiv resolution as of 17 September 2026.

5 Provenance

The mathematics was produced without human intervention in three stages. (1) The result was found by gpt-6-astra, reasoning effort max, in a single pass begun on 2026-09-17, for catalog id 1602.05184__00 in attacks_retry. It was a second attempt: the prompt supplied a gpt-5.6-sol attempt proving only the universal-vertex and a co-bipartite special case. The new dominated-vertex reduction closes the remaining case; inherited material is not presented as new. (2) An adversarial claude-opus-5 referee audited every step and ran the recorded computations on 2026-09-17; its verdict was MINOR GAPS. (3) On 2026-09-17 claude-fable-5-1 created only a recoverable note skeleton before its session ended. On 2026-09-18 Codex completed the present rewrite, reorganised the argument, and moved full proofs to [Appendix A](#). This version explicitly repairs every named gap: it explains why two distinct

shortest paths contribute two extra edges; treats the predecessor class containing z when z has no same-layer neighbour; obtains a leaf outside $\{p, q\}$ using the prescribed tree edge pq ; attributes (1) and the good-edge terminology; cites the earlier deletion lemma; and records the He manuscript. No other mathematics was changed. **No human mathematician has checked this argument.**

A Full proofs

Lemma 2.1 (Dominated-vertex deletion). *Suppose G is 2-connected, $|V(G)| \geq 4$, and $N_G[u] \subseteq N_G[v]$ for distinct u, v . Put $H = G - u$. Then:*

- (i) H is isometric in G ;
- (ii) the only possible cut vertex of H is v ;
- (iii) if $q_G(u)$ counts incidences $(\{a, b\}, e)$ in which $a, b \in V(H)$ and e is incident with u and good for $\{a, b\}$, then $\eta(G) - \eta(H) = c_G(u) + q_G(u)$;
- (iv) $q_G(u) = 0$ when u is simplicial, while $q_G(u) \geq 2$ otherwise.

Proof. Any portion $x - u - y$ of a path between vertices of H can be replaced by a walk through v of length at most two (if $x = v$, then $xy \in E(G)$). Thus deleting u preserves all distances in H . If $w \neq v$ were a cut vertex of H , every vertex of $N_G(u) \setminus \{w\}$ would lie in the component of $H - w$ containing v . Restoring u could not join the other components, contradicting connectivity of $G - w$.

Since H is isometric, an edge of H is good for an old pair in G exactly when it is good in H . The additional contributions are pairs containing u , totalling $c_G(u)$, and old-pair incidences counted by $q_G(u)$. If u is simplicial, take the last neighbour y of u on a shortest $a-u$ path. Every $x \in N(u)$ is adjacent to y , so $d(a, x) \leq d(a, u)$; no edge ux is good for a pair avoiding u . If u is not simplicial, nonadjacent $x, y \in N(u)$ make both xu, uy good for $\{x, y\}$. $\square$

Lemma 2.2 (Low-contribution structure). *Let G be 2-connected and let a be non-universal with $c_G(a) \leq 3$. Every vertex lies at distance at most two from a , and one of the following holds, where $A = N(a)$.*

- I. *There are adjacent $p, q \in A$, a vertex $z \notin N[a]$ with $N(z) \cap A = \{p, q\}$, and a set X such that $V(G) \setminus N[a] = \{z\} \dot{\cup} X$. Either X is empty, or $G[X]$ is connected, every $x \in X$ has $N(x) \cap A = \{p\}$, and exactly one edge joins z to X .*
- II. *There are distinct $p, q \in A$ and nonempty X, Y partitioning $V(G) \setminus N[a]$ such that $G[X], G[Y]$ are connected, $N(x) \cap A = \{p\}$ for $x \in X$, $N(y) \cap A = \{q\}$ for $y \in Y$, and exactly one edge xy joins X to Y .*

Moreover every non-universal vertex of a 2-connected graph satisfies $c_G(a) \geq 2$.

Proof. Use distance layers L_i from a ; a neighbour in L_{i-1} is a predecessor. Let M be the vertices having more than one shortest path from a . If $b \in M$, the union of two distinct shortest $a-b$ paths has at least $d(a, b) + 2$ edges, all good for $\{a, b\}$. A union of only $d(a, b) + 1$ edges would make one path replace an edge of the other by a parallel edge, impossible in a simple graph. Thus $D_G(a, b) \geq 2$ and $|M| \leq 1$. This supplies a justification requested by the referee.

If $x, y \in L_i$ have unique shortest paths, distinct predecessors, and $xy \in E(G)$, then $d(x, p(y)) = 2$ and $p(y)y$ is extra good for $\{a, x\}$; symmetrically $p(x)x$ is extra for $\{a, y\}$. Different neighbours yield different extra edges.

Case $M = \emptyset$. Edges between consecutive layers form a rooted tree. Let F be same-layer edges whose endpoints have different predecessors. Then $c_G(a) \geq 2|F|$, so $|F| \leq 1$. If $F = \emptyset$, proper descendants of any non-root vertex have no exit except through it, contradicting 2-connectivity unless the tree has depth one; then a is universal. Thus $F = \{xy\}$, with predecessors p, q .

The vertices p, q have a common predecessor. Otherwise $pq \notin E(G)$ and they have no common neighbour: one in L_{i-2} contradicts distinct predecessors, one in L_{i-1} creates another edge of F , and one in L_i creates two shortest paths. Hence $d(p, q) = 3$ via $p - x - y - q$, while $d(p, y) = d(q, x) = 2$; edges qy, px add two further contributions and force $c_G(a) \geq 4$. Let w be the common predecessor. The exceptional edge xy lies among descendants of w , so 2-connectivity forces $w = a$, and $i = 2$. The same separation argument forbids later layers. Any third predecessor class, or a component of the p -class avoiding x (and symmetrically for q, y), is separated by its predecessor. This is Type II.

Case $M = \{z\}$. The vertex z has at least two predecessors and no successor. With $r \geq 3$ predecessors, the last two layers of its shortest paths contain $2r$ good edges, only two on one fixed path, so $D_G(a, z) \geq 2r - 2 \geq 4$. Thus z has exactly two predecessors p, q . They are adjacent: otherwise qz is extra for $\{a, p\}$ and pz for $\{a, q\}$, on top of $D_G(a, z) \geq 2$.

There is no same-layer edge between uniquely reached vertices of distinct predecessor classes. In particular p, q have a common predecessor w . A same-layer neighbour t of z must have predecessor p or q , or both pz, qz are extra for $\{a, t\}$; even then it contributes one, so z has at most one such neighbour.

Choose a breadth-first tree using p as predecessor of z . All exceptional edges lie among descendants of w , so $w = a, z \in L_2$, and no later layer exists. If z has no same-layer neighbour, any other predecessor class is separated by its predecessor. In its own class C_p , z is isolated; any other vertex lies in a component of $G[C_p]$ avoiding z and is separated by p . Thus $L_2 = \{z\}$. *This repairs the first gap named by the referee.* If z has the sole neighbour t , take p as predecessor of t . Every remaining L_2 vertex has predecessor p , and their induced graph is connected and contains t ; otherwise p separates a component. This is Type I.

Finally, $c_G(a) \leq 1$ forces $M = F = \emptyset$, which makes a universal. Hence every non-universal a has $c_G(a) \geq 2$. $\square$

Proposition 2.3 (Four-unit reduction). *Let G be 2-connected, have no universal vertex, and satisfy $\eta(G) < 2|V(G)|$. Then either $G \cong C_5$, or there is a vertex u dominated by another vertex such that $G - u$ is 2-connected and $\eta(G) - \eta(G - u) \geq 4$.*

Proof. Equation (1) gives a with $c_G(a) \leq 3$. Apply Lemma 2.2. We first find a dominated vertex with 2-connected deletion.

In Type I, if $X = \emptyset$, z is simplicial with adjacent neighbours p, q ; if $X = \{t\}$, t is simplicial with adjacent neighbours p, z . Deleting a simplicial vertex from a 2-connected graph of order at least four preserves 2-connectivity. If $|X| \geq 2$, choose $u \in X \setminus \{t\}$ with $G[X] - u$ connected. Then u is dominated by p , and $G - \{u, p\}$ is connected through the sole edge tz ; by Lemma 2.1(ii), $G - u$ is 2-connected.

In Type II the same works if $|X| \geq 2$ or $|Y| \geq 2$, choosing a non-cut vertex away from the cross-edge endpoint. Thus let $X = \{x\}, Y = \{y\}$. If $A = \{p, q\}$, then $pq \notin E(G)$ gives C_5 , while $pq \in E(G)$ makes a simplicial. If $|A| \geq 3$, $R = G[A] + pq$ is connected. Choose a spanning tree T containing pq . Removing pq splits T into T_p, T_q ; one has at least two vertices and thus a leaf u distinct from p, q . Hence $R - u$ is connected. The vertex u is dominated by a , and replacing pq by $p - x - y - q$ shows $G - \{a, u\}$ is connected. Thus $G - u$ is 2-connected. *This use of $T - pq$ repairs the second gap named by the referee.*

If the resulting u is non-simplicial, Lemma 2.2 gives $c_G(u) \geq 2$ and Lemma 2.1 gives $q_G(u) \geq 2$. Suppose u is simplicial and $c_G(u) \leq 3$. Apply Lemma 2.2 to u . Type I would make p universal. In Type II, $p - x - y - q - p$ is an isometric C_4 and contributes 8. Every $b \in (\{u\} \cup N(u)) \setminus \{p, q\}$ has $D_G(b, x), D_G(b, y) \geq 1$. For $b \in X \setminus \{x\}$, if bx is an edge, xy is extra for $\{b, q\}$ and pq for $\{b, y\}$; otherwise the two shortest paths $b - p - q - y, b - p - x - y$ span five good edges and $D_G(b, y) \geq 2$. The Y side is symmetric. These pairs are distinct, so $\eta(G) \geq 8 + 2(|V(G)| - 4) = 2|V(G)|$, a contradiction. Therefore a simplicial u has $c_G(u) \geq 4$. In either case Lemma 2.1(iii) gives a loss at least four. $\square$

Proposition 3.1 (Universal-vertex estimate). *If G is 2-connected of order n , has a universal vertex, and is not K_n , K_n^2 , K_n^{n-2} , then $\eta(G) \geq 3n - 10$.*

Proof. If $\text{diam}(G) \leq 2$, then for a nonedge ab , every common neighbour supplies the two good edges of a length-two path and every further good edge is the middle edge of an induced P_4 with endpoints a, b . For an edge ab , each good edge other than ab is opposite it in an induced C_4 . Thus

$$\eta(G) = 2(p_3(G) - \overline{m}(G)) + p_4(G) + 4c_4(G). \quad (2)$$

For $G = K_1 \vee Q$, the apex creates one P_3 for each nonedge of Q and lies in no induced P_4, C_4 , so

$$\eta(G) = 2p_3(Q) + p_4(Q) + 4c_4(Q). \quad (3)$$

Choose a maximum clique C of connected noncomplete Q , of order $c \geq 2$, and distance layers L_i from C . Put $\ell = |L_1|$, $r = \sum_{i \geq 2} |L_i|$. If $x \in L_1$ has a_x neighbours in C , then $a_x(c - a_x) \geq c - 1$ induced P_3 's contain x . Every later-layer vertex gives another P_3 and a P_4 from a shortest path (for L_2 , append a nonneighbour in C of its predecessor). The unique maximum-layer vertex distinguishes the copies. Hence

$$p_3(Q) \geq (c - 1)\ell + r, \quad p_4(Q) \geq r.$$

For $c \geq 3, \ell \geq 2$, $2(c - 1)\ell + 3r$ exceeds $3|Q| - 7$ by $(2c - 5)\ell - 3c + 7 \geq c - 3$.

If $c \geq 3, \ell = 1$, write $L_1 = \{x\}$ and $A = |N(x) \cap C|(c - |N(x) \cap C|) \geq c - 1$. If $r = 0$, the value is $2A$; only $K_{|Q|} - e$ and a clique with a pendant vertex fail $3|Q| - 7$. If $r > 0$, put $s = |L_2|$, $t = \sum_{i \geq 3} |L_i|$. Each L_2 vertex gives A induced P_4 's, and

$$2p_3 + p_4 + 4c_4 \geq 2A + (A + 2)s + 3t \geq 3|Q| - 7.$$

If $c = 2$, say $C = \{v, w\}$, split L_1 into $A_0 = N(v) \cap L_1, B_0 = N(w) \cap L_1$ of sizes α, β , and let e count edges between them. Pairs inside A_0, B_0 give extra P_3 's; nonedges between them give P_4 's and edges give C_4 's. Therefore

$$\begin{aligned} 2p_3 + p_4 + 4c_4 &\geq 2\ell + 3r + \alpha(\alpha - 1) + \beta(\beta - 1) + \alpha\beta + 3e \\ &\geq 3\ell + 3r - 1 = 3|Q| - 7. \end{aligned}$$

The two exceptions have value $2|Q| - 4$. Since $|Q| = n - 1$, (3) gives $3n - 10$ unless the cone is K_n, K_n^{n-2} , or K_n^2 . $\square$

Theorem 1.2 (Main theorem). *Let G be a 2-connected graph of order $n \geq 4$, with $G \not\cong K_n, K_n^2, K_n^{n-2}$. Then*

$$\eta(G) \geq \min\{2n, 3n - 10\}.$$

Both branches are sharp for every applicable $n \geq 5$.

Proof. Equation (2) gives

$$\eta(K_m^t) = 2(m - 1 - t)(t - 1), \quad (4)$$

so $\eta(K_m^2) = \eta(K_m^{m-2}) = 2m - 6$. Put $f(n) = \min\{2n, 3n - 10\}$ and induct on n .

At $n = 4$, the only admissible graph is C_4 , and $\eta(C_4) = 8 \geq f(4)$. Let $n \geq 5$. A universal vertex is handled by Proposition 3.1. Without one, if $\eta(G) \geq 2n$ we are done, and $\eta(C_5) = 5 = f(5)$. Otherwise Proposition 2.3 gives dominated u such that $H = G - u$ is 2-connected and $\eta(G) \geq \eta(H) + 4$. Set $m = n - 1$. If H is not exceptional, induction gives

$$\eta(G) \geq f(n - 1) + 4 = \min\{2n + 2, 3n - 9\} \geq f(n).$$

If $H = K_m$, every neighbour of u is universal in G , impossible. If $H = K_m - e$ with missing edge xy , all other vertices are universal in H and hence nonadjacent to u . Since G is 2-connected, $N_G(u) = \{x, y\}$, but neither dominates u , a contradiction.

Suppose $H = K_m^2$; $m = 4$ is the preceding case. Write H as a clique B of order $m - 1$ plus z with $N_H(z) = \{p, q\}$, and put $R = B \setminus \{p, q\}$. Since p, q are universal in H , u is adjacent to neither. Nor is u adjacent to z : it would need a second neighbour in R , and no neighbour of u could dominate both z and that vertex. Hence $N_G(u) \subseteq R$. Put $t = |N_G(u)| \geq 2$. Then u is simplicial. The t common neighbours give $D_G(u, p), D_G(u, q) \geq 2(t - 1)$. Also $d_G(u, z) = 3$, and the paths $u - r - p - z, u - r - q - z$ over $r \in N(u)$ have $3t + 2$ distinct good edges, so $D_G(u, z) \geq 3t - 1$. Thus $c_G(u) \geq 7t - 5 \geq 9$, whence

$$\eta(G) \geq (2m - 6) + 9 = 2n + 1,$$

contrary to the branch assumption. This completes the induction.

For sharpness, (3) gives $\eta(K_1 \vee P_{n-1}) = 3n - 10$. In the clique-plus-matched-edge family of Section 4, start with C_4 and add $n - 4$ simplicial clique vertices, each contributing two; Lemma 2.1 gives $\eta = 8 + 2(n - 4) = 2n$. $\square$

B Report of the LLM referee (verbatim)

Status of this appendix

The following report is reproduced verbatim as an audit trail. Its references to the original writeup predate the explicit repairs and citations listed in Section 5.

B.1 Interpretation

The writeup proves exactly Conjecture 5 of Bonamy–Knor–Lužar–Pinlou–Škrekovski (arXiv:1602.05184, Appl. Math. Comput. 312 (2017) 202–213), with no weakening or literal-reading gamesmanship. I verified against the ar5iv full text of the source paper:

- Source definition: “let K_n^t be the graph obtained from K_{n-1} by adding one new vertex adjacent to t of the $n-1$ old vertices” — identical to the writeup’s §1.
- Source Conjecture 5: “Let G be a 2-connected graph of order $n \geq 10$ not isomorphic to K_n , K_n^2 , or $K_n^{\{n-2\}}$. Then $\eta(G) \geq 2n$ ” — identical to what is proved.
- $\eta(G) = \text{Sz}(G) - W(G)$ with the standard $\text{Sz}(G) = \sum_{\{uv \in E\}} n_{uv}(u) - n_{uv}(v)$ — identical.

The writeup in fact proves something strictly stronger and for all $n \geq 4$: $\eta(G) \geq \min\{2n, 3n-10\}$, i.e. $\eta \geq 3n-10$ for $n \leq 10$ and $\eta \geq 2n$ for $n \geq 10$. My exhaustive computation shows this stronger form is *exactly* tight at every order $5 \leq n \leq 11$ (minima 5, 8, 11, 14, 17, 20, 22), so the formulation is not accidentally weak, and the $n = 10$ transition in the conjecture is genuine. A reasonable author of the original problem would consider the conjecture resolved (modulo the two writing gaps below).

This is an `attacks_retry` target. The previous attempt (`gpt-5.6-sol`, verdict `partial`) proved the universal-vertex case only. The new writeup reproduces that part (as its §5, essentially verbatim) and adds the genuinely new content: the dominated-vertex reduction of §§2–4. I checked the reproduced §5 independently rather than crediting it; it is correct (one exposition improvement over the original: the distinctness of the counted induced subgraphs is now justified by “unique vertex of maximum layer”). No error flagged in the earlier attempt is reproduced — the earlier attempt’s own honest statement of what remained open (“no step above proves that a graph with $\eta(G) < 2n$ must belong to either treated class”) is precisely what §§2–4 now supply.

B.2 Step-by-step findings

Numbering: §n.k = my enumeration of the writeup’s claims in order.

step	label	note
1. Theorem (1): $\eta \geq \min\{2n, 3n-10\}$, $n \geq 4$, $G \notin \{K_n, K_n^2, K_n^{\{n-2\}}\}$		Conclusion “ $n \geq 10 \Rightarrow \eta \geq 2n$ ” is right: $3n-10 \geq 2n \iff n \geq 10$. Exhaustively tight for $5 \leq n \leq 11$ (see Computational check).
2. Identity (2): $\eta(G) = \sum_{\{\text{pairs}\}} D_G(a,b) = \frac{1}{2} \sum_{a \in V} \sum_{b \in V} D_G(a,b)$ VALID (attribution gap)		Correct: for edge $e = uv$, the pairs $\{a,b\}$ for which e is good number exactly $n_{uv}(u) - n_{uv}(v)$, so $\sum_{\{\text{pairs}\}} D_G(a,b) = \sum_{uv \in E} (n_{uv}(u) - n_{uv}(v)) = \text{Sz}(G) - W(G)$. But this is Proposition 6 of the source paper , and the “good edge” notion is credited there to Simić–Gutman–Baltić (Math. Slovaca 50 (2000) 1–15). The writeup presents it as its own “double counting” with no citation.
3. Lemma 2.1(1): $N[u] \subseteq N[v] \Rightarrow G-u$ isometric	VALID	Any $x-u-y$ detour is replaced by $x-v-y$ (or $x=v$, then $xy \in E$), length ≤ 2 .

step	label	note
4. Lemma 2.1(2): only possible cut vertex of $H = G - u$ is v	VALID	All of $N_G(u) \setminus \{w\}$ lies in v 's component of $H - w$, so re-adding u cannot reconnect $G - w$.
5. Lemma 2.1(3): $\eta(G) - \eta(H) = c_G(u) + q_G(u)$	VALID	Exact, by isometry: pairs inside H keep their distances, their good edges inside H are unchanged, the only extra good edges are those incident with u (counted by q), plus the pairs containing u (counted by c). Machine-verified for every dominated pair in every 2-connected graph on ≤ 9 vertices (194,066 graphs) and on an 820k sample at $n = 10$.
6. Lemma 2.1(4): u simplicial $\Rightarrow q = 0$; u non-simplicial $\Rightarrow q \geq 2$	VALID	Simplicial: $d(a, x) \leq d(a, u)$ for all $x \in N(u)$, so ux is never good for a pair avoiding u . Non-simplicial: for nonadjacent $x, y \in N(u)$, both xu and uy are good for $\{x, y\}$. Machine-verified as above.
7. Deleting a simplicial vertex from a 2-connected graph of order ≥ 4 keeps 2-connectivity	VALID	The given one-line argument is correct; machine-verified for $n \leq 9$.
8. Lemma 3.1, (5): b has ≥ 2 shortest a - b paths $\Rightarrow D(a, b) \geq 2$, hence $ M \leq 1$	VALID	The union of two distinct shortest a - b paths has $\geq d+2$ edges: if it had only $d+1$, the second path would use $d-1$ edges of the first plus one new edge parallel to the missing one — impossible in a simple graph. The writeup asserts “ $\geq d(a, b)+2$ ” without this remark; it is correct but worth spelling out.
9. Lemma 3.1, same-layer observation: $xy \in E$, $x, y \in L_i$ uniquely reached with distinct predecessors $\Rightarrow p(y)y$ is an extra good edge for $\{a, x\}$	VALID	$d(x, p(y)) = 2$ because x has a unique predecessor $\neq p(y)$; the edge is off the unique shortest a - x path; distinct y give distinct edges.
10. Lemma 3.1, Case 1 ($M = \emptyset$), $F = \emptyset \Rightarrow a$ universal	VALID	With $M = \emptyset$ every non-tree edge is a sibling edge, so proper descendants of any non-root tree vertex w see only T_w ; 2-connectivity forces depth 1.
11. Lemma 3.1, Case 1: $F = \{xy\} \Rightarrow p = p(x)$, $q = p(y)$ have a common predecessor	VALID	Otherwise $pq \notin E$ and p, q have no common neighbour (all three sub-cases checked), so $d(p, q) = 3$ and qy, px are extra good edges for $\{a, p\}$, $\{a, q\}$: $c \geq 4$.
12. Lemma 3.1, Case 1: $w = a$, $i = 2$, no L_3 , Type II	VALID	Subtree-separation argument is correct; class-separation forces exactly the two classes X (at p) and Y (at q), each connected, with the single cross edge xy .
13. Lemma 3.1, Case 2 ($M = \{z\}$): z has exactly 2 predecessors p, q and $pq \in E$	VALID	$r \geq 3$ gives $\geq 2r$ good edges in the last two layers, of which one fixed shortest path uses 2, so $D(a, z) \geq 2r - 2 \geq 4$. $pq \notin E$ gives pz, qz as extra good edges for $\{a, q\}$, $\{a, p\}$, so $c \geq 4$. Such an edge adds ≥ 1 at each endpoint on top of $D(a, z) \geq 2$. Subtree separation then forces $w = a$, $z \in L_2$, no L_3 .
14. Lemma 3.1, Case 2: no same-layer edge between uniquely reached vertices with different predecessors; p, q have a common predecessor $w = a$	VALID	
15. Lemma 3.1, Case 2: z has ≤ 1 same-layer neighbour; conclusion Type I	GAP (one line, repairable)	
16. Lemma 3.1, (4): a non-universal $\Rightarrow c_G(a) \geq 2$	VALID	The bounds are right. But “if z has no same-layer neighbour, every <i>other</i> predecessor class in L_2 would be separated by its predecessor, so $L_2 = \{z\}$ ” silently ignores z 's own class C_p : one must add that z is then isolated inside $G[C_p]$, so any other vertex of C_p lies in a component of $G[C_p]$ avoiding z and is separated by p . With that sentence the argument is complete.
17. Lemma 4.1, Type I ($X = \emptyset$, $X = \{t\}$, $ X \geq 2$)	VALID	$c \leq 1$ forces $M = \emptyset$ and $F = \emptyset$, hence a universal by step 10. Machine-verified for all $n \leq 9$ plus the $n = 10$ sample.
18. Lemma 4.1, Type II with $ X \geq 2$ (or $ Y \geq 2$)	VALID	z (resp. t) is simplicial with adjacent neighbours; for $ X \geq 2$ the chosen u is a non-cut vertex of $G[X]$ other than t , dominated by p , and $G - \{u, p\}$ is connected ($X \setminus \{u\}$ connected and hooked to z by tz ; $\{a\} \cup (A \setminus \{p\}) \cup \{z\}$ connected through a and q). Lemma 2.1(2) finishes. Same argument, cross edge xy .

step	label	note
19. Lemma 4.1, Type II, $X = \{x\}$, $Y = \{y\}$, $A = \{p, q\} \Rightarrow C_5$ or a simplicial a	VALID	$n = 5$ here; $pq \notin E$ gives exactly C_5 , $pq \in E$ makes a simplicial and dominated by p.
20. Lemma 4.1, Type II, $X = \{x\}$, $Y = \{y\}$, $ A \geq 3$: GAP (one line, repairable) leaf of a spanning tree of $R = G[A] + pq$		“Since this tree has at least three vertices, it has a leaf $u \notin \{p, q\}$ ” is false as a general statement about trees (a path from p to q has only p, q as leaves). It is nevertheless true here because the tree was chosen to contain the edge pq: deleting that edge splits it into T_p, T_q , one of which has ≥ 2 vertices and therefore supplies a leaf of T outside $\{p, q\}$. The writeup must say this. The rest (u dominated by a, $G - \{a, u\}$ connected via $p-x-y-q$) is correct.
21. Lemma 4.2: Type I impossible when a is simplicial	VALID	p would be adjacent to a, $A \setminus \{p\}$, z and all of X, i.e. universal.
22. Lemma 4.2: $p-x-y-q-p$ is an induced isometric C_4 , contributing $\geq \eta(C_4) = 8$	VALID	A is a clique so $pq \in E$; $py, qx \notin E$ by the Type II neighbourhood condition; all six pairwise distances agree with C_4 , and isometry makes every C_4 -good edge good in G. $\eta(C_4) = 16 - 8 = 8$ re-derived.
23. Lemma 4.2: each $b \in K \setminus \{p, q\}$ gives $D(b, x)$, $D(b, y) \geq 1$	VALID	$d(b, x) = d(b, y) = 2$ and qy (resp. px) is a good edge off the shortest path.
24. Lemma 4.2: each $b \in X \setminus \{x\}$ (and $Y \setminus \{y\}$) gives 2 more units	VALID	$bx \in E$: xy is extra for $\{b, q\}$ and pq is extra for $\{b, y\}$. $bx \notin E$: $d(b, y) = 3$ with two shortest paths spanning 5 good edges, so $D(b, y) \geq 2$.
25. Lemma 4.2 total: $\eta \geq 8 + 2(n-4) = 2n$	VALID	The counted pairs are pairwise distinct (one endpoint inside the 4-cycle, one outside) and all other $D \geq 0$. Machine-verified as a statement for all $n \leq 9$ and the $n = 10$ sample.
26. Prop 4.3: reduction with increment ≥ 4	VALID	$\eta < 2n \Rightarrow \Sigma c < 4n \Rightarrow$ some $c(a) \leq 3$; Lemma 4.1 gives a dominated u with $G-u$ 2-connected; non-simplicial u: $c \geq 2$ (step 16) and $q \geq 2$ (step 6); simplicial u: Lemma 4.2 applied to u gives $c(u) \geq 4$ and $q = 0$. Note the lemma is correctly applied to an arbitrary simplicial vertex, not only to the low-contribution vertex a. Machine-verified (existence of a ≥ 4 increment) for all $n \leq 9$ and the $n = 10$ sample.
27. Lemma 5.1: $\text{diam} \leq 2 \Rightarrow \eta = 2(p_3 - \bar{m}) + p_4 + 4c_4$	VALID	Re-derived: for a non-edge ab the good edges are $2 \cdot N(a) \cap N(b) $ plus the middle edges of induced P_4 's with ends a, b; for an edge ab the only extra good edge is the opposite edge of an induced C_4 , each C_4 being counted by its 4 edges. Machine-verified on every 2-connected graph of diameter ≤ 2 with $n \leq 9$, and on all connected graphs up to $m = 8$ via the cone construction.
28. (12): $\eta(K_1 \vee Q) = 2p_3(Q) + p_4(Q) + 4c_4(Q)$	VALID	$p_3(G) = p_3(Q) + \bar{m}(Q)$, $\bar{m}(G) = \bar{m}(Q)$, and the apex lies in no induced P_4/C_4 . Machine-verified for all connected Q with $m \leq 8$ (12,109 graphs).
29. Lemma 5.2, Case 1 ($c \geq 3$, $\ell \geq 2$)	VALID	$F \geq 2(c-1)\ell + 3r$; difference to $3m-7$ is $(2c-5)\ell - 3c + 7 \geq c-3 \geq 0$ (re-derived).
30. Lemma 5.2, Case 2 with $r = 0$ (the two exceptional graphs)	VALID	$F = 2a(c-a)$ exactly; $a = 1$ and $a = c-1$ are the exceptions with $F = 2m-4$; otherwise $4(c-2) \geq 3c-4$ for $c \geq 4$, and $2 \leq a \leq c-2$ already forces $c \geq 4$.
31. Lemma 5.2, Case 2 with $r > 0$	VALID	$p_3 \geq A+s+t$, $p_4 \geq As+t$ (the A s copies $y-x-v-u$ are induced since L_2 has no C-neighbours); difference $2A + (A-1)s - 3c + 4 \geq 0$ using $s \geq 1$, $A \geq c-1$.
32. Lemma 5.2, Case 3 ($c = 2$, triangle-free)	VALID	A, B partition L_1 (no triangle); the extra P_3 's, the $\alpha\beta$ -e induced P_4 's and the e induced C_4 's are all distinct from the layer copies; $\alpha(\alpha-1) + \beta(\beta-1) + \alpha\beta \geq \alpha + \beta - 1$ checked in both sub-cases; $F \geq 3\ell + 3r - 1 = 3m - 7$.
33. Prop 5.3: universal vertex $\Rightarrow \eta \geq 3n - 10$	VALID	$Q = G - \text{apex}$ is connected; the two Lemma 5.2 exceptions are exactly $K_n \setminus \{n-2\}$ and $K_n \setminus 2$. Machine-verified for all cones over connected Q with $m \leq 8$.

step	label	note
34. (17): $\eta(K_m^t) = 2(m-1)(t-1)$, so $\eta(K_m^2) = \eta(K_m^{m-2}) = 2m-6$	VALID	Re-derived from (11); machine-verified for all $4 \leq m \leq 12$ and $1 \leq t \leq m-2$. Agrees with the source paper.
35. §6 base case $n = 4$	VALID	The only 2-connected non-exceptional graph on 4 vertices is C_4 (geng confirms 3 such graphs: K_4 , K_4-e , C_4); $\eta(C_4) = 8 \geq f(4) = 2$. $\min\{2n+2, 3n-9\} \geq \min\{2n, 3n-10\}$.
36. §6 Case 1 arithmetic: $f(n-1) + 4 \geq f(n)$	VALID	Any neighbour of u would be universal in G .
37. §6 Case 2: $H \cong K_m$ impossible	VALID	$N_G(u) = \{x, y\}$ with $xy \notin E$, so no neighbour of u dominates u , contradicting the choice of u .
38. §6 Case 3: $H \cong K_m - e$ impossible	VALID	Re-derived all three inequalities: $D(u, p)$, $D(u, q) \geq 2(t-1)$ (t common neighbours, $d = 2$); $d(u, z) = 3$ with $3t+2$ good edges so $D(u, z) \geq 3t-1$; $c(u) \geq 7t-5 \geq 9$ for $t \geq 2$; $\eta \geq (2m-6) + 9 = 2n+1$. The exclusion of $u \sim z$ and $u \sim p, q$ is correct, and $t \geq 2$ needs $ R \geq 2$, i.e. $m \geq 5$, which is exactly what Case 3 leaves.
39. §6 Case 4: $H \cong K_m^2$ gives $\eta(G) \geq 2n+1$, contradiction	VALID	Verified by code for $5 \leq n \leq 15$: 5,8,...,35 and 10,12,...,30 respectively, both 2-connected and non-exceptional.
40. §7 sharpness: $\eta(K_{1 \vee P_{n-1}}) = 3n-10$ and $\eta(\text{clique } K_{n-2} + \text{edge } xy \text{ joined by a } 2\text{-matching}) = 2n$	VALID	

No step is labelled ERROR. The two GAP entries (steps 15 and 20) are single missing sentences whose repairs I supply above and verified myself; neither threatens the result.

B.3 Reference check

The writeup is deliberately self-contained and cites nothing, so there is no miscitation to flag. I nevertheless checked everything it implicitly leans on, via the arXiv full text of arXiv:1602.05184:

- **Conjecture 5** and the definition of K_n^t : quoted above, identical to the writeup's reading. ✓
- **Proposition 6** of the source paper is exactly the writeup's identity (2) ($\eta = \sum_{\text{pairs}}(g - d)$); the source attributes the "good edge" notion to Simić et al. [15] (Math. Slovaca 50 (2000) 1–15). The writeup should cite this rather than derive it anonymously. This is an attribution gap, not a correctness problem. ✓/flag
- **Theorem 2** ($\eta \geq 2n-6$ for 2-connected non-complete) and **Theorem 4** ($\eta \geq 2n-5$ after excluding K_n^2 , K_n^{n-2}) of the source paper are both implied by, and consistent with, the writeup's $\min\{2n, 3n-10\}$ (e.g. $n = 5$: $5 \geq 5$; $n = 6$: $8 \geq 7$). ✓
- **Lemma 7** of the source paper ("if $G-u$ is 2-connected, not complete, and $N[u] \subseteq N[v]$, then $\eta(G) - \eta(G-u) \geq 2$ ") is the weaker ancestor of the writeup's Lemma 2.1(3); the writeup's exact version (increment = $c + q$) is stronger and is what makes the ≥ 4 increment possible. Not cited, again an attribution matter. ✓/flag

Priority check (step 2b/2c). A dedicated literature sweep (OpenAlex citing set of the source paper — 13 works, Semantic Scholar — 15, arXiv API for all "Szeged" abstracts since 2024, zbMATH, and the Knor–Škrekovski–Tepoh survey arXiv:2303.11405, whose full text I had grepped) found **no published proof or disproof of Conjecture 5**. It remains open in the refereed and arXiv literature as of 2026-09-17. One item deserves explicit mention:

- **Zhijie He, "The Strengthened Szeged–Wiener Conjecture"**, a manuscript posted 2026-09-08 (nine days before this attack ran) at github.com/FDmd233/strengthened-szeged-wiener-conjecture and submitted as a solution to MathDB problem 332514. Its Theorem 1 is verbatim Conjecture 5, and its method (minimum counterexample + dominated-vertex induction, the K_n^t formula, exceptional 2-connected deletions, plus a computer certificate at order 10) overlaps the writeup's strategy. **However:** it has no arXiv identifier, no DOI, no venue, is indexed nowhere, and MathDB's own raw status line still reads "Open. A reader-submitted proof claims to settle the strengthened inequality, but it is incomplete and unverified, so the problem remains open." It is not this campaign's own artifact (different author, different repository, predates the attack), so it is not self-contamination — but by the standard of step 2c (an independent paper with authors, a date **and a venue or arXiv id**) it does not qualify as prior art. I therefore do **not** mark ALREADY_KNOWN; I flag it as a live priority risk that a human should check before any submission, and note that if that manuscript is correct, the writeup's contribution is independent rediscovery, not priority.
- No published source contains the diameter- ≤ 2 identity of Lemma 5.1 in the $p_3/p_4/c_4$ form (the nearest published diameter-2 Szeged formulas, Ilić arXiv:1103.2513 Prop. 3.1/3.3, are Zagreb-index expressions under an extra "every edge in exactly t triangles" hypothesis — a different statement). So Lemma 5.1 appears to be new, and true.

B.4 Computational check

All scripts are in `verification_astra/scripts/1602.05184__00/`. I re-ran (not merely inherited) everything I rely on. Two independent implementations of η were used (a C brute force `eta.c/eta2` and a Python one in `check_lemmas.py`), and they agree; the Python code additionally cross-checks $\eta = Sz - W$ against $\eta = \sum D(a, b)$ on every graph.

A. The theorem itself, exhaustively (nauty geng -c, all 2-connected graphs):

n	2-connected graphs	min η over non-exceptional	$\min\{2n, 3n-10\}$	violations
4	3	8	2	0

n	2-connected graphs	min η over non-exceptional	$\min\{2n, 3n-10\}$	violations
5	10	5	5	0
6	56	8	8	0
7	468	11	11	0
8	7,123	14	14	0
9	194,066	17	17	0
10	9,743,542	20	20	0
11	900,969,091	22	22	0

I also decoded the minimisers: at $n = 10$ the value 20 is attained by a graph with a universal vertex (the cone $K_1 \vee P_9$, the $3n-10$ family), and at $n = 11$ the value 22 is attained by graph6 JQ11\Jnz|~_, degree sequence $[2,2,8,8,8,8,8,8,9,9]$ — exactly the writeup’s §7 family (clique K_9 plus an edge xy joined to it by a 2-matching). So both sharpness families in §7 are the genuine extremal graphs at the relevant orders.

The minimum equals the claimed bound **exactly** at every order 5–11: the theorem is sharp, not merely true, and the $3n-10 \rightarrow 2n$ crossover at $n = 10$ is real (at $n = 11$, $3n-10 = 23$ but the true minimum is 22). The $n = 11$ row is my own re-run (72 shards, `rerun11.sh`), reproducing the earlier referee run’s figures exactly.

B. Every lemma of the writeup, mechanically, on all small 2-connected graphs (`check_lemmas.py`: checks the identity (2), Lemma 2.1(3),(4) for every dominated pair, simplicial-deletion 2-connectivity, Lemma 3.1’s $c \geq 2$ and the full Type I / Type II structure classification for every vertex with $c \leq 3$, Lemma 4.1’s existence of a dominated vertex with 2-connected deletion, Lemma 4.2, Prop 4.3’s ≥ 4 increment, Lemma 5.1, and the theorem):

- $n = 5,6,7,8$: 10 / 56 / 468 / 7,123 graphs — **ALL CHECKS PASSED**.
- $n = 9$: all 194,066 graphs (48 shards) — **ALL CHECKS PASSED**.
- $n = 10$: a uniform 8.4% sample, 820,139 graphs (48 of 2000 `geng` shards) — **ALL CHECKS PASSED**.

C. Section 5 separately (`check_sec5.py`, all connected graphs Q , cone identity (12), Lemma 5.2’s bound and its two exceptional values, Prop 5.3): $m = 4\dots 8$, 6 / 21 / 112 / 853 / 11,117 graphs — **ALL PASSED**.

D. Sharpness (`check_sharpness.py`): $\eta(K_m^t) = 2(m-1-t)(t-1)$ confirmed for all $4 \leq m \leq 12$, $1 \leq t \leq m-2$; $\eta(K_1 \vee P_{n-1}) = 3n-10$ and $\eta(\text{clique } K_{n-2} + xy \text{ attached by a 2-matching}) = 2n$ confirmed for $5 \leq n \leq 15$, both families 2-connected and non-exceptional.

E. Counterexample hunt beyond the exhaustive range ($n \geq 12$) — the main thing the earlier referee run had not finished. Nothing below $2n$ was found anywhere:

1. *Simulated annealing* on η over 2-connected graphs (`anneal.c`). Validated first: it recovers the exact exhaustive minima 17, 20, 22 at $n = 9, 10, 11$. Then 5 seeds $\times$ 40 restarts $\times$ 60,000 steps for $n = 12, 13, 14, 15, 16, 17, 18, 20, 22, 24$: best non-exceptional η found was exactly $2n$ for $n = 12, 13, 15, 17, 18, 20, 22, 24$ and above $2n$ for 14,16; **0 graphs with $\eta < 2n$** .
2. *Exhaustive over all graphs containing a large clique* (`bigclique.c`; enumerates every graph whose first $n-k$ vertices form a clique, up to permutation of the clique part, so it covers **all** graphs with clique number $\geq n-k$):
 - $n = 11, k = 5$ (clique ≥ 6): 2,221,702,531 graphs, min $\eta = 22 = 2n$, **0 below**.
 - $n = 12, k = 4$ (clique ≥ 8): 29,741,514 graphs, min $\eta = 24 = 2n$, **0 below**.
 - $n = 13, k = 4$: 80,657,066 graphs, min $\eta = 26 = 2n$, **0 below**.
 - $n = 14, k = 4$: 203,901,610 graphs, min $\eta = 28 = 2n$, **0 below**.
 - $n = 12, k = 5$ (clique ≥ 7): 1,182,743,608 graphs in the 22 of 32 shards completed at the time of writing, min $\eta = 24 = 2n$, **0 below**. (I stopped the remaining 10 shards after ~22 minutes; the work is heavily skewed towards them, and no shard produced a single violation. Re-runnable with `bigclique 12 5 <shard> 32`.)
3. *Structured families* (`families.py`), $n = 12\dots 30$: cliques with attached paths of length 1–8, cliques plus two vertices with all attachment patterns of size ≤ 5 (adjacent and non-adjacent), two cliques joined by 2–4 cross edges, complete multipartite graphs, cycle powers C_n^k , clique plus two internally disjoint paths. Minimum over everything is exactly $2n$, attained only by the writeup’s §7 family; **0 counterexamples**.
4. *Exhaustive perturbations* of both extremal families at $n = 12, 13$ (`perturb3.py`): all 1-, 2- and 3-edge flips, 35,924 / 42,778 / 60,850 / 71,773 resulting 2-connected non-exceptional graphs; minimum $\eta = 24 (= 2n)$ at $n = 12$ and 26 ($= 2n$) at $n = 13$ around the clique+matching family, 26 and 29 ($= 3n-10 > 2n$) around the cone family; **0 below $2n$** .

F. A caution about a contrary claim. My literature sub-search reported, from random sampling, that the true minimum for $n \geq 10$ is $4n-16$ (attained by K_n^3) and that $2n$ is therefore not tight. That is **wrong**, and my exhaustive data refutes it: at $n = 11$ the minimum over all 900,969,091 2-connected graphs is $22 = 2n$, whereas $4n-16 = 28$. Random sampling simply never hits the sparse extremal family (a clique K_{n-2} plus an edge xy joined to it by a 2-matching), which attains exactly $2n$ for every $n \geq 5$. The writeup’s sharpness section is correct and the conjectured constant $2n$ is best possible.

B.5 Caveats

1. **Two one-line justification gaps** (steps 15 and 20 above): the $L_z = \{z\}$ argument in Lemma 3.1 Case 2 does not treat z ’s own predecessor class, and “a tree with ≥ 3 vertices has a leaf outside $\{p, q\}$ ” is false in general and needs the observation that the spanning tree was chosen to contain the edge pq . Both are repaired in one sentence each; I verified both repairs.
2. **Missing attribution.** Identity (2) is Proposition 6 of the very paper that poses the conjecture, and the “good edge” concept is due to Simić–Gutman–Baltić; Lemma 2.1 is a sharpened form of the source paper’s Lemma 7. A submitted version must cite these. The writeup’s closing claim “no computational enumeration or unproved external assertion is used” is literally true and is a real strength — the proof is fully self-contained — but it should not be read as a claim that the framework is original.
3. **Priority risk.** An unrefereed manuscript claiming exactly this theorem by a similar route was posted publicly nine days before this attack (Zhijie He, 2026-09-08, GitHub + MathDB 332514). Not indexed, no venue, and marked “incomplete and unverified” by the host, so I do not treat it as prior art — but it must be checked by a human before any submission.

4. **Scope statements.** The theorem is proved for $n \geq 4$; the conjecture's hypothesis $n \geq 10$ is only used at the very end ($3n-10 \geq 2n$). The three exceptional graphs are excluded exactly as in the source. Edge cases are handled: $n = 4$ base case, C_5 as the one sporadic obstruction in Lemma 4.1/Prop 4.3, $m = 4$ folded into Case 3 of §6, and $t \geq 2$ in Case 4 needing $m \geq 5$ (which is exactly what Case 3 leaves open). The empty and disconnected cases do not arise since G is 2-connected throughout.
5. **What is *not* claimed.** The writeup does not characterise the extremal graphs for $\eta = 2n$ (my searches suggest the clique $K_{\lfloor n-2 \rfloor} + 2$ -matching family is the only one, but this is not proved and not asserted), and does not extend to 1-connected graphs.
6. **Residual verification limits.** The lemma-level machine check is exhaustive only to $n = 9$ (plus an 8.4% sample at $n = 10$); the theorem is exhaustive to $n = 11$; beyond that the evidence is a very large but not exhaustive search. The correctness argument at large n rests on the hand audit above, which found no error.

B.6 Referee summary

I set out to break this writeup and could not. The interpretation is exactly the source paper's Conjecture 5, and the claimed theorem is strictly stronger ($\eta \geq \min\{2n, 3n-10\}$ for all $n \geq 4$) yet *exactly* tight at every order $5 \leq n \leq 11$ according to my own exhaustive enumeration — including all 900,969,091 2-connected graphs on 11 vertices, where the minimum is $22 = 2n$ and nothing falls below. The proof's genuinely new part, the dominated-vertex reduction of §§2–4, survives a line-by-line audit: the low-contribution structure lemma (Lemma 3.1) correctly extracts a rigid Type I / Type II shape from $c(a) \leq 3$, Lemma 4.1 correctly produces a dominated vertex whose deletion stays 2-connected, Lemma 4.2 correctly blocks the cheap simplicial case, and Prop 4.3's increment of at least 4 is exactly what the induction needs; all of these were also verified mechanically on every 2-connected graph up to 9 vertices and on an 820k sample at 10 vertices, with zero failures. The inherited §5 (universal-vertex case) was re-checked independently and is correct, as are the §6 exceptional-predecessor cases and the §7 sharpness families. The specific counterexample hunt the earlier referee run had begun — graphs beyond the exhaustive range — turned up nothing: annealing (validated against the known minima), exhaustive enumeration of all graphs containing a clique of order $n-4$ at $n = 12, 13, 14$ and $n-5$ at $n = 11, 12$ (over 3.4 billion graphs in total), structured families to $n = 30$, and all ≤ 3 -edge perturbations of the extremal graphs at $n = 12, 13$ all bottom out at exactly $2n$. I therefore record MINOR_GAPS rather than CONFIRMED only because two justifications are incomplete as written (a missing sentence about z 's own predecessor class in Lemma 3.1 Case 2, and the false-as-stated “a tree with ≥ 3 vertices has a leaf outside $\{p, q\}$ ” in Lemma 4.1, which holds only because the spanning tree was chosen to contain pq) and because the identity (2) is Proposition 6 of the source paper, used without citation. A human should also check the unrefereed manuscript posted on GitHub / MathDB on 2026-09-08 that claims the same theorem by a similar route before any submission.

References

- [1] M. Bonamy, M. Knor, B. Lužar, A. Pinlou and R. Škrekovski, *On the difference between the Szeged and Wiener index*, Applied Mathematics and Computation **312** (2017), 202–213; [arXiv:1602.05184](https://arxiv.org/abs/1602.05184).
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- [3] Z. He, *The Strengthened Szeged–Wiener Conjecture*, unrefereed manuscript posted 8 September 2026, <https://github.com/FDmd233/strengthened-szeged-wiener-conjecture>.